

Technical Document #569: Cloud Computing - Comprehensive Overview

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Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Serverless computing allows developers to run code without managing servers.
- Containerization packages applications with their dependencies.
- Platform as a Service (PaaS) provides a platform for developing and deploying applications.